15/3/26

Re-referencing

To measure the height, we need a base. Similary to measure the potential we need a reference. we already know, in EEG the signals are recorded based on potential difference b/w 2 electrode. The reference electrode need to be placed at a point where there is less electrophysical activity (eg:- earlobe, mastoid, nose tip etc..). All places in human body have some degree of electrical activity, hence choose a reference point in the body based on Project requirement.

Strategy

1. Single electrode → earlobe, mastoid
2. Linked ear → connect left & right ear
3. Average → use average of all recorded electrode
4. REST → Ref Electrode standardization Technique

Re-referencing is also one of the preprocessing steps in EEG data collection.

EEG artifact.

The signal collected from scalp is not a clean signal, it always has some disruption in the signal (due to various factors like eye movement, eye blinking, environmental factors), which is called artifact.

- EEG Artifact
  - Physiological artifact — Body activity
  - Non-physiological Artifact — External factors

Physiological activity

- Eye blink artifact → High amplitude, Narrow spike pattern
  - → Max amplitude at FP1, FP2, FPZ
- Eye movement artifact (EOG) Electro ocular artifact. → Large amplitude & wide frequency range
- Electromyographic artifact (EMG) → due to facial muscle movement
  - → Sharp freq, high peak for short duration.
- ECG artifact → Caused due to electrical activity of the heart
- Motion artifacts → Non periodic waveform
- Sweat artifact → low freq, Low amplitude.

Non physiological activity

- ↳ caused due to no strong contact b/w electrode & skin
- ↳ Much harder to identify. due to unpredictable pattern.

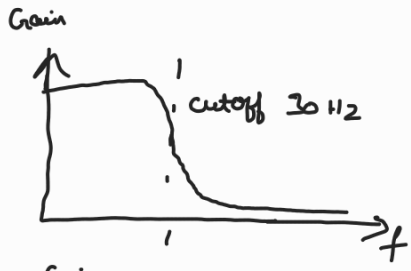
① Electrical artifact (due to electromagnetic around us laptop, charger etc...)

② Electrode artifact (poor contact of electrode - occur in specific electrode, not in a pattern)

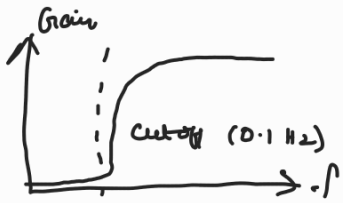
Filtering EEG signal

Sometimes signal may get interrupted by high power line interference (50 - 60 Hz) which may lead to noise in the signal, To improve the quality of signal we use filter, which is like a gate keeper which decides which frequencies are allowed to pass through, and which are blocked.

① Low pass filter → Allows low freq to pass



② High pass filter → Allows high freq filter



③ Band pass filter → Allows specific freq of signal

④ Notch filter → Blocks specific frequency signal